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DEPARTMENT OF DEFENSE HANDBOOK

AIRCRAFT NUCLEAR COMPATIBILITY CERTIFICATION

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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the United States Air Force and the Department of Energy (DOE) National Nuclear Security Administration (NNSA), and Sandia National Laboratory (SNL).

2. This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply.

3. This Handbook describes the electrical and mechanical responsibilities for the tasks related to Aircraft Compatibility Nuclear Certification.

4. The content of this Handbook elaborates on the aircraft compatibility certification tasks described in AFI 63-125, Nuclear Certification Program; MIL-STD-1822, Nuclear Compatibility Certification of Nuclear Weapon Systems, Sub-Systems, and Support Equipment; and SAND2011-2981, Aircraft Compatibility Tasks Required for the Release of an Aircraft Compatibility Control Drawing. This Handbook is written to help the reader understand all of the electrical and mechanical responsibilities associated with aircraft compatibility, and the criteria that measures the process elements leading to a successful and approved Nuclear Compatibility Certification Statement (NCCS), a released aircraft Compatibility Control Drawing (ACCD), and/or Compatibility Certification (CC) Drawing.

5. Comments, suggestions, or questions on this document should be addressed to the Defense Standardization Program Office (DSPO), 1551 Wyoming Blvd. SE Kirtland AFB, NM 87117-5624 or emailed to afnwc.dspo.workflow@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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1. SCOPE

The information in this Handbook describes a way of learning and dealing with the aircraft nuclear compatibility certification tasks required by the Air Force Nuclear Weapons Center (AFNWC), NNSA, and SNL. This handbook provides guidance for the requirements defined in MIL-STD-1822 (Section 4), and provides a description in paragraph 4.21 how ground launched missile compatibility certification is accomplished for comparison purposes.

To structure the information in this Handbook, the tasks are defined in AFI 63-125 and MIL-STD-1822. This Handbook will help the reader understand what the job/task is and what needs to be done to complete the aircraft compatibility certification task. The Air Force Nuclear Certification Process is described in detail in AFI 63-125. The Major elements of nuclear certification are: design certification and operational certification. Components of design certification are: compatibility certification, nuclear safety design certification, safety rules development, and technical order certification. Although all components of design certification are closely related and arguably difficult to totally separate, this document will focus on mechanical and electrical nuclear weapon compatibility certification on the aircraft platform. The Nuclear Weapon System Compatibility is the capability of two or more nuclear weapon system components of equipment or material to exist or function in the same system or environment without mutual interference. Nuclear weapon system compatibility evaluations are a component of Operational Suitability as established in AFI 63-101/20-101. Nuclear weapon system operational suitability is a measure of the degree to which a nuclear weapon system or end-item can be placed satisfactorily into field use with consideration given to availability, compatibility, transportation, interoperability, and reliability. A successful nuclear weapon system compatibility evaluation (and subsequent NCCS) is a component of the Air Force's Nuclear Weapon System Certification and Systems Engineering processes. Nuclear safety design certification requirements are closely related to compatibility requirements. They are described in detail in AFMAN 91-118, Safety Design and Evaluation Criteria for Nuclear Weapon Systems, and AFMAN 91-119, Safety Design and Evaluation Criteria for Nuclear Weapon Systems Software, and are not repeated here. According to DODD 3150.02, section 4.2.8, "Nuclear weapon system safety, security, survivability, and use control are interrelated. Decisions concerning one should not be made without consideration of the effect of those decisions on the others." Guidance for evaluation and reporting for nuclear compatibility certification are also delineated.

This Handbook also covers the needs of survivable nuclear weapons need survivable support equipment in paragraph 4.6. To address the interrelation of survivability with safety and security, AFNWC has a nuclear technology analysis branch, which provides scientific oversight of the RN part of the CBRN (Chemical, Biological, Radiological, Nuclear) survivability mission. This branch monitors development and modification of nuclear weapons, their components, or subassemblies for compliance with Military Characteristics and Stockpile-To-Target Sequence survivability requirements.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in this Handbook.

2.2 Government Documents. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

MILITARY STANDARDS

MIL-STD-1822	Nuclear Compatibility Certification of Nuclear Weapon systems, Sub-Systems, and Support Equipment

(Copies of this document are available online at https://assist.dla.mil)

MEMORANDUM OF AGREEMENT

MOU No. DE-GM04-2001AL77133	Agreement on Division of Responsibilities for Aircraft Monitor and Control (AMAC) Systems Design Requirements and Compatibility Testing

(Copies of these document are available from the Air Force Nuclear Weapons Center (AFNWC) Nuclear Technology (NT) Directorate, 1551 Wyoming Blvd SE, Bldg 20332, Kirtland AFB, NM 87117

SANDIA NATIONAL LABORATORIES

SAND92-0828	F-15E Nuclear Certification/Compatibility/ Testing and Evaluation
SAND91-2664	B-52H Nuclear Certification Tasks Performed by the Aircraft Compatibility Department at Sandia National Laboratories
SAND94-2845	Sandia's Aircraft Compatibility Activities on the F-16 Aircraft
SAND99-0400	Sandia's Electrical Aircraft Compatibility Activities on the B-2A Aircraft
SAND97-3147	System 1 AMAC Design Handbook for Weapon System Designers and the AMAC Project Officers Group
SAND2011-2981	Aircraft Compatibility Tasks Required for the Release of an Aircraft Compatibility Control Drawing (ACCD) (U)

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SANDIA NATIONAL LABORATORIES

SAND92-1527	Sandia's Aircraft Compatibility Activities on the Tornado PA-200 Aircraft

(Copies of these documents are available from the Sandia National Laboratory Technical Library, amparso@sandia.gov).

SPECIFICATION STANDARD NO.

SYS 1001	AMAC POG System 1
SYS 2001	AMAC POG System 2
SYS 1300	System 1 Aircraft Monitor and Control Test Requirements
SYS 2300	System 2 Aircraft Monitor and Control Test Requirements

(Copies of these document are available from the Air Force Nuclear Weapons Center (AFNWC) Nuclear Technology (NT) Directorate, 1551 Wyoming Blvd SE, Bldg 20332, Kirtland AFB, NM 87117

AIR FORCE INSTRUCTIONS (AFI)

AFI 63-125	Nuclear Certification Program

(Copies of this document are available on line at http://www.e-publishing.af.mil)

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3. DEFINITIONS AND ACRONYMS

3.1 Definitions. The definitions used in this handbook are defined as follows:

3.1.1 Aircraft Compatibility Control Drawing (ACCD). Each nuclear weapon is issued a Major Assembly Release (MAR). The part of the MAR that describes aircraft compatibility is called the ACCD. The ACCD establishes the extent of compatibility and restrictions between the nuclear weapon and the aircraft.

3.1.2 Compatibility Certification Drawing (CC). A control drawing prepared and maintained by Sandia which establishes the extent of compatibility and restrictions between a nuclear warhead on an Air Launched Cruise Missile and an aircraft. The CC is released after the compatibility tasks have been successfully completed. The CC is maintained by Sandia.

3.1.3 Major Assembly Release (MAR). A Sandia-prepared, NNSA-approved statement that DOE stockpile weapon material is satisfactory for release on a designated effective date to the DOD for specified capabilities and uses that are qualified by limitations and exceptions to the MC and STS requirements. Compatibility must be evaluated with each of the weapon systems designated by the STS or MC. The ACCD (gravity bombs) and the CC (warhead) are subsets of the MAR. All nuclear weapons have a MAR.

3.1.4 Military Characteristics (MC). The MC is a document that defines the using service's requirements for a specific nuclear bomb/warhead. It describes the required weapon yields and fusing options, warhead parameters and general information concerning operational, physical, functional, environmental, vulnerability, safety, reliability, maintenance, monitoring, storage and handling considerations, and sets forth the design compliance priorities in the event of conflicting design requirements.

3.1.5 Nuclear Compatibility Certification Statement (NCCS). The NCCS is a document that is written for each aircraft platform that has been certified to carry nuclear weapons. It defines general information about the aircraft system, Nuclear Certified Equipment (NCE), and approved software. The NCCS also defines the nuclear weapon system configuration, carriage/delivery parameters, any restrictions, and certification and surveillance AMAC tests conducted to date. The NCCS is kept up-to-date and is released in accordance with AFI 63-125, Nuclear Certification Program.

3.1.6 Stockpile-to-Target Sequence (STS). The STS is a service generated document that defines the operational and logistical concepts and related physical environments involved in delivering a nuclear weapon from the stockpile to the target. It also defines the logistical flow involved in moving and recycling nuclear material to and from the user to stockpile. The STS furnishes the NNSA with a set of design requirements that amplify the MC's by providing additional detailed guidance necessary for development of a nuclear weapon. The STS provides detailed information in such areas as stockpile storage temperatures and durations, logistic movement, vibration and shock environments, specific aircraft carriage configurations, aircraft carriage sound pressure levels, soak temperatures, delivery profiles, etc.

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3.1.7 Master Nuclear Certification List (MNCL). The MNCL is a web-based tool designed to define the hardware, software, technical data, configuration, and other information that is encompassed within the definition of nuclear certification defined in AFI 63-125. Its purpose is to enable users to identify nuclear certification status of a weapon system, sub-system, component, software, or support equipment.

3.2 Acronyms. The acronyms used in this handbook are defined as follows:

AC	Alternating Current
A/C	Aircraft
ACCD	Aircraft Compatibility Control Drawing
AF&F	Arming, Fuzing, and Firing
AFI	Air Force Instruction
AFMAN	Air Force Manual
AFNWC	Air Force Nuclear Weapons Center
ALCM	Air Launch Cruise Missile
AMAC	Aircraft Monitor And Control
AMC	Alternate Mission Equipment
APU	Auxiliary Power Unit
A/V	Air Vehicle
BDU	Bomb Dummy Unit
CAD	Computer Aided Design
CC	Compatibility Certification
CD	Command Disable
CFD	Computational Fluid Dynamics
Cg	Center of gravity
Config	Configuration
CM	Cruise Missile
CONOPS	Concept of Operations
CPIN	Computer Program Identification Number
CSRL	Common Strategic Rotary Launcher
CTU	Compatibility Test Unit

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DC	Direct Current
DoD	Department of Defense
DOE	Department of Energy
DOF	Degrees of Freedom
ECM	Electronic Counter Measure
EICD	Electrical Interface Control Drawing
Elec	Electrical
EM	ElectroMagnetic
EMC	ElectroMagnetic Compatibility
EMMS	ElectroMagnetic Measurement System
Equip	Equipment
ESDD	Electrical System Description Drawing
FASTER	Functional AMAC System Tester
FDAR	Final Design Approval Report
FTU	Flight Test Unit
FWSD	Full Weapon System Demonstration/Drop
GND	Ground
Hz	Hertz
HQ	Headquarters
ICD	Interface Control Drawing/Document
IMF	Integrated Maintenance Facility
JTA	Joint Test Assembly
LEP	Life Extension Program
M&S	Modeling and Simulation
MAR	Major Assembly Release
MC	Military Characteristics
Mech	Mechanical
MICD	Mechanical Interface Control Drawing
MNCL	Master Nuclear Certification List

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MOD	Modified
MOU	Memorandum of Understanding
NATO	North American Treaty Organization
NCCS	Nuclear Compatibility Certification Statement
NCE	Nuclear Certified Equipment
NIIN	National Item Identification Number
NNSA	National Nuclear Security Administration
PAL	Permissive Action Link
PDR	Preliminary Design Report
PLS	Prearm Load Simulator
POG	Project Officer's Group
POM	Project Officer Meeting
Prelim	Preliminary
Prod	Production
PSD	Power Spectral Density
PSICD	Platform Store Interface Control Document
q	Dynamic pressure
RA	Responsible Agency
RND	Rules Need Date
SA	Supporting Agency
SIL	System Integration Laboratory
SM	Single Manager
SNL	Sandia National Laboratories
SOC	Statement Of Compatibility
SPO	System Program Office
SRAM	Short Range Attack Missile
Struct	Structural
STS	Stockpile-to-Target Sequence
SWIFT	Special Weapons Interface Tester

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T/M	Telemetry
TO	Technical Order
UR	Unsatisfactory Report
USAF	United States Air Force
USG	Unique Signal Generator or Generation
VV&A	Verification, Validation, & Accreditation
VATU	Vibration Acoustic Test Unit
VFA	Vibration Fly Around Assembly
WCD	Weapon Capability Date
WPN	Weapon (Bomb or Cruise Missile)
WR	War Reserve

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4. BACKGROUND

4.1 General Background. The AFNWC at Kirtland AFB, NM and the SNL Aircraft Compatibility Department are made up of staff members with electrical, software, and mechanical backgrounds. The Nuclear Technology and Interagency Directorate of the AFNWC have two main Sections within the Surety and Effect Branch which are responsible for aircraft compatibility activities. The Nuclear Surety Section has staff assigned to handle, aircraft compatibility, Aircraft Monitor and Control (AMAC) testing, nuclear safety, and nuclear weapon-related Technical Orders (TOs). The Certification Management Section handles nuclear certification activities as defined in Figure 1.1 of AFI 63-125. Within the Sandia Aircraft Compatibility Department, an aircraft is assigned to an electrical engineer and a mechanical engineer (hereafter referred to as the Sandia engineer). SNL is a Prime Contractor for the NNSA and is chartered to ensure that capability exists between US nuclear weapons and the aircraft they are carried on throughout their life in the inventory. The AFNWC and SNL team members that conduct the various aircraft compatibility tests, work together to ensure both the United States Air Force (USAF) and NNSA that a capability exists between the weapon and the aircraft for all nuclear weapons carried on the aircraft. This responsibility exists for the entire life of both the aircraft and the weapons. The AFNWC and NNSA agreement on division of responsibilities are defined in Memorandum of Understanding (MOU) DE-GM04-2001AL77133 (Sections VI and VII). The MOU delineates the responsibilities regarding the design requirements, test requirements, and documentation of AMAC used with aircraft-delivered nuclear weapons (bombs and cruise missiles) developed by the NNSA with overall weapon system requirements established by the DoD. The AFNWC and Sandia Aircraft Compatibility Department work with the various military agencies and aircraft contractors to assure that any given aircraft has a capability to successfully monitor, control, and deliver the nuclear weapon(s) it is required to carry. Part of this process involves attending aircraft and weapon Project Officer Meetings (POMs), and conducting various mechanical and electrical tests. These meetings and tests are attended by agencies that have some level of responsibility for nuclear certification of the aircraft or weapon. Sandia and the AFNWC work together to coordinate compatibility certification requirements into the aircraft program so that a capability can be eventually granted and maintained during the life of the aircraft.

Sandia Aircraft Compatibility works with Sandia's Air Delivered Weapons Systems Engineering Centers throughout the life of the weapon program to assure an electrical and mechanical capability exists between each weapon and the delivery aircraft. The Aircraft Compatibility Department keeps these Centers up to date on changes to the AMAC specifications that defines the aircraft/weapon interface and on changes to the aircraft's AMAC system and delivery characteristics that might impact the weapon's ability to reliably function. The Weapon Departments within these Centers, in turn, keep the Aircraft Compatibility Department and the AFNWC updated on what changes have occurred, or will occur because of a new development or stockpile improvement on a given weapon system. This is the process by which the gravity weapon ACCD and the cruise missile CC documents are kept current by SNL, and in turn the aircraft NCCS documents are kept current by the AFNWC.

The ACCD is a drawing that exists on each nuclear gravity bomb in the inventory that describes in detail the compatible configurations on any given aircraft with that bomb. This includes such things as a list of compatible aircraft, compatible AMAC controllers, allowed

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mixed weapon load configurations, allowed configurations of Electronic Counter Measure (ECM) pods, delivery limitations, and compatible AMAC software with the bomb. The ACCD exists as long as the bomb is in the inventory and is maintained by the weapon departments at Sandia (SAND 2011-2981).

The ACCD equivalent for cruise missile warheads carried on aircraft is the CC drawing. The SNL Aircraft Compatibility Department is responsible for the compatibility aspects of the W80 nuclear warhead on an air launched cruise missile.

4.1.1 Primary participants/roles. The Compatibility Certification process requires significant involvement throughout the development community. Participants include, but are not limited to: (1) the Single/System Program Managers; (2) AFNWC/NT or NI; (3) Sandia National Laboratories (SNL); (4) the lead and using Major Commands (MAJCOMs); and when needed, (5) the Air Force SEEK EAGLE Office (AFSEO) and a variety of test and evaluation organizations including contractors.

4.1.1.1 Single/System Program Manager. To obtain Compatibility Certification for aircraft, the DoD Single/System Program Manager generates the required Compatibility Certification documents such as th Nuclear Certification Impact Statement, Electrical Interface Control Drawing (EICD), Mechanical Interface Control Drawing (MICD), Preliminary Design Report (PDR), Final Design Approval Report (FDAR), and appropriate technical orders as specified in the Certification Requirements Plan. For cruise missile and ICBM systems, the Single/System Program Manager is responsible for updating appropriate interface documents.

4.1.1.2 AFNWC/NT or NI. If the Certification Requirements Plan identifies requirements for testing and analysis to complete the Compatibility Certification process, then the AFNWC/NT (for aircraft systems) or the NI (for ICBM systems) will coordinate the necessary testing, demonstration and analysis with the appropriate test organizations, and the National Nuclear Security Administration, as needed. The AFNWC/NT issues or updates the Nuclear Compatibility Certification Statement for aircraft systems and the ICBM NI is responsible for the compatibility certification of ICBM systems. For aircraft systems, the AFNWC and SNL will determine the AMAC testing necessary for each aircraft system. The AFNWC/NT will conduct and direct testing as required. The results of AMAC testing are maintained in a database. These results are published in a test report following each test and used as source data justifying nuclear compatibility certification. The AFNWC/NT provides AMAC test results to the system's Single/System Program Manager and briefs these results at the weapon delivery system's Nuclear Project Officers Meeting (NPOM).

4.1.1.3 Sandia National Laboratories (SNL). The aircraft compatibility department at SNL has the responsibility to assure that the aircraft/cruise missiles that have been designed to carry and release nuclear weapons are indeed compatible with the nuclear bombs/warheads. SNL personnel work with the Air Force and their contractors to ensure that electrical, mechanical, and aerodynamic compatibility certification requirements are fully understood, and are full participants in conducting required compatibility certification testing, demonstrations, and analyses. At the conclusion of an aircraft/cruise missile compatibility certification program, SNL updates the

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ACCD for aircraft or the CC Drawing for cruise missiles, for inclusion into the Major Assembly Release (MAR). The MAR is also used for the ground launched missile warheads.

4.1.1.4 Lead/Using MAJCOM. The Lead Command is the MAJCOM that serves as the operators' interface with the Single Manager for a weapon system (i.e., Air Combat Command for the F-15, Air Force Global Strike Command for the B-2). The Using Commands are the MAJCOMs that operate the weapon system, subsystem or equipment item (i.e., USAFE for F-16s and F-15s). HQ USAF designates operational commands or organizations authorized to conduct or participate in operations or operational testing. In addition to all other testing and analysis, the responsible MAJCOM determines if a system requires further testing of the release system with the weapon delivery vehicle. If flight clearance or safe separation test and analysis are required, the MAJCOM may require further testing in conjunction with testing conducted by the AFSEO. Results of the tests are provided to the appropriate System Program Office.

4.1.1.5 Air Force SEEK EAGLE Office (AFSEO). The Air Force SEEK EAGLE certification program determines safe upload and download procedures, flight limits for safe carriage, employment, jettison, safe separation, and ballistic accuracy for all nuclear weapons and for associated suspension equipment, tanks, and pods carried internally or externally on the aircraft. The AFSEO will initiate updates to appropriate aircraft technical orders and flight manuals as a result of their activities. Recertification is required for any change which alters the aerodynamic characteristics of the aircraft or store, and ejection characteristics of the suspension equipment.

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5. COMPATIBILITY CERTIFICATION ACTIVITIES

The description in this section of compatibility certification activities identifies which agencies are considered as Responsible or Supporting. Specific compatibility certification requirements are defined in MIL-STD-1822. The AFI 63-125 Certification Requirements Plan (CRP) defines these activities for new and existing weapon systems undergoing modification in order to achieve nuclear certification. Cruise Missiles and Ground Launched Missiles are subject to Compatibility Certification activities similar to those of an aircraft/bomb with the exception that a CC is developed in place of the ACCD for Cruise Missiles, and the MAR for Ground Launched Missiles.

5.1 Six (6) Degrees of Freedom (6 DOF) Model.

5.1.1 Develop 6 DOF model of nuclear weapon system and conduct system analysis using computer simulation (Responsible Agency (RA): Single Manager/Contractors, Supporting Agencies (SA): AFNWC/SNL). Modern day high-speed computers, combined with proven Modeling and Simulation (M&S) techniques, allow very complex system interactions to be accurately represented throughout the range of expected operations. Common techniques consider 6 DOF models that are analyzed through finite-element/matrix analysis. The 6 DOF model analyses allow potential critical operations to be identified and aids in the development of follow-on wind tunnel and separation test planning.

5.1.2 Verification, Validation, & Accreditation (VV&A) of 6 DOF Model (RA: SNL, SA: AFNWC). Throughout the process of the nuclear weapon system's compatibility certification, actual test results will be compared to M&S results that may result in changes to assumptions and techniques employed to ensure the model accurately represents the system throughout the spectrum of expected operations. The VV&A process ensures: (1) Verification that the computer simulation/programming yields expected results, (2) Validation that the model adequately represents the actual system, and (3) Accreditation that the user has a high confidence in using the model results. Once the 6 DOF model has undergone the VV&A process, the model may be used in lieu of actual wind tunnel and separation tests to conduct post-compatibility certification system analyses.

5.2 Wind Tunnel Tests (RA: Single Manager (SM), SA: SNL/Contractors/AFNWC).

Wind tunnel tests are normally performed to obtain: (1) store loads data and (2) separation data prior to conducting flight tests. The USAF is interested in obtaining data associated with loads imposed on the aircraft by the store while SNL is most interested in information about the loads imposed on the bombs/warheads. The key to successful wind tunnel testing is to get involved early in an aircraft development program when the aircraft contractor is conducting wind tunnel testing for conventional stores. At this point the costs are minimal and the facilities and test hardware are available. The wind tunnel tests are generally performed in DoD facilities with the DoD paying for tests involving new aircraft or cruise missiles, and the DOE paying for tests on new bombs/warheads or modifications to existing ones. Things to consider during Wind Tunnel Tests are:

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5.2.1 Internal Carriage. The "Doors Closed" loads environments have been measured but have never been large enough to merit an analysis. The doors-open and doors- opening-and-closing transients are the critical flight conditions that need loads analyses. Sandia aerodynamicists in the past have stated that internal carriage loads data is very difficult to obtain because of scaled-down weapons and weapons bays and the interference of wind tunnel instrumentation hardware. Critical loads, if required, are usually obtained during full-scale development flights for Lug and Sway Brace Loads Analysis. Separation of scaled weapons out of weapons bays can be satisfactorily accomplished. Various photo techniques showing pitch, yaw, and roll at selected ejection velocities are used to provide data.

5.2.2 External Carriage. Wind tunnel testing is expensive but ideally suited to provide external carriage loads and separation data early in the aircraft and/or weapon development phase. Various store loading configurations can be tested leaving the "worst case" loadings to be flight tested. Wind tunnel time provides better comparative data than an aircraft flight test because tunnel conditions can be very accurately controlled. Flight test conditions are frequently compromised due to outside factors which cannot always be counted on to provide precise flight profiles. Wind tunnels have a variety of methods and techniques to obtain loads and separation data. The Responsible Agency consults with Sandia's Aerosciences Department to select the one best method suited to obtain the data needed, and then coordinates with the using service and the aircraft contractor to join in on their tests. Weapon models and wind tunnel data support are provided by Sandia's Engineering Sciences Center. It is also important to get the Sandia Aircraft Compatibility Department involved as the wind tunnel test data supports the required loads analyses on the nuclear weapons required for conducting Vibration Fly-Around tests and analysis. Wind tunnel testing is many times done in conjunction with the Air Force and SNL's Aerosciences, Aircraft Compatibility, and Wapon System Engineering Departments.

5.3 Lug and Sway Brace Loads Analysis (RA: SM, SA: Contractors/AFNWC/SNL). The purpose of the lug and sway brace analysis program is to determine the mechanical loads imposed on weapon lugs and on the weapon case where the lugs attach and where the nuclear weapon rack sway braces make contact during carriage. The loads are determined for the full flight envelope of the aircraft. The flight envelope is a definition of Mach number or speed versus altitude requirements for the aircraft. The nuclear weapon is designed to a set of requirements based on aircraft type called out in the original Military Characteristics (MCs) and this analysis verifies that the aircraft being added to the requirements (if that is the case) does not impose higher loads on the lugs and on the weapon case. Although it is possible to directly measure these loads by using instrumented nuclear weapon test units, this has been found to be prohibitively expensive and so other means of producing the required data have been developed. Nuclear bomb caseloads are generally predicted using 6 DOF modeling and simulation based on the aerodynamic characteristics, which have been measured in wind tunnel tests, and the performance characteristics of the aircraft. The aircraft performance characteristics are generally available from the aircraft development agency.

Sandia's Aircraft Compatibility group in concurrence with Weapon Systems Engineering and the Structural Analysis Department use the Lug and Sway brace Analysis (LUGSAN) software to compute the lug and sway brace loads resulting from various aircraft maneuvers. This software program requires data that defines the maximum performance capability of the aircraft,

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the associated bomb aerodynamic drag coefficients, geometric data about the store-aircraft interface, and the location of the Center of Gravity (Cg) of the store with respect to the Cg of the aircraft. This data is obtained from wind tunnel tests, MICDs, liaison with the aircraft contractor, etc.

The lug and sway brace loads analysis in an aircraft/nuclear weapon compatibility program is required before starting the Vibration Fly-Around (VFA) tests. This analysis can be tailored to the approximate flight profile required by the test and is used to satisfy the services safety of flight requirements. Upon completion of the VFA tests, a final loads analysis is performed before the capability is added to the ACCD or CC. If the analysis indicates limitations exists, those limitations are to be stated in the ACCD and CC notes.

As an example of gathering loads data on an aircraft, the B-2 aircraft compatibility program took a different approach to store loads. Northrop looked at aircraft geometric data, ejector rack sway brace torque, g-loadings on the aircraft, and other factors and came up with a preliminary loads analysis. Sandia studied the data and determined that the combined loads were below the strength capabilities of the B61 and B83, and determined that it was okay to fly the scheduled vibration fly-arounds. Following this analysis, the B-2 Combined Test Force was tasked to record loads data from instrumented B61 and B83 dummy shapes during some of their doors-open carriage demonstrations. The next step was to integrate the measured flight loads into the preliminary loads analysis, and produce a final loads analysis report. Sandia requested a flight test with a bomb in the 3 or 9 o'clock position on the rotary launcher, with lug and sway brace instrumentation, and then eject the bomb that would be in the 6 o'clock position. The high ejector velocities required on this aircraft, result in high compression loads on the lower sway brace pads and high loads in tension on the bomb lugs of those bombs in the 3 or 9 o'clock position. So depending upon the stores load configuration on an aircraft, it may require consideration of accomplishing loads data on adjacent stations, while another store is being released.

5.4 Static Ejection Tests (RA: SM, SA: Contractors/AFNWC/SNL). Static ejection tests are conducted for initial compatibility and whenever questions arise about the magnitude of shock spectra that will be imposed on nuclear weapon components during ejection. A newly designed or redesigned rack will undergo static ejection testing. When an ejector cartridge's pneumatic performance is changed, the modified performance will require static ejection testing to confirm that shock and vibration limits of the nuclear weapon components are not exceeded. Static ejection tests can also be used to help determine the nuclear weapon pitch and ejection velocity that will occur during weapon release. The test configuration generally consists of a DoD supplied rack mounted on a relatively rigid beam with either a SNL-supplied Static Ejection Test Unit (SETU) loaded on the rack or the W80 Vibration Acoustic Test Unit (VATU) loaded in a cruise missile that is loaded on the rack. These units are specially designed test units containing components that have been instrumented to measure the shock spectra for later analysis. The unit is ejected off the rack and the acceleration data spectra are recorded. A video of the test is usually recorded using a high speed camera to record pitch angle and velocity information. The tests may be run at a contractor facility or at a DoD engineering facility. These tests may also be run with the rack mounted on a real aircraft in order to get real-world (less severe) results.

The net result in a cartridge change will only change the force applied by the ejector pistons to the outside weapon case. On ejector racks with dual pistons there is usually a mechanical

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device in the cartridge gas plumbing system that will allow some throttling of the gas to a forward or aft piston, thus allowing the test agency to change the pitch of the bomb at release. Prior to the first nuclear bomb release from a new aircraft, the static weapon pitch should duplicate the weapon pitch and ejection velocity recorded during wind tunnel tests. For example, the rack orifices on the MAU-12C/A rack on a USAF aircraft for internal carriage were changed part way through the flight testing to accomplish a smaller nose-down pitch at higher q conditions.

An interesting problem always shows up when one attempts to do a series of static ejection tests: what to use to absorb the shock of the test unit being ejected. If the shock-absorbing material is too soft, it bottoms out; if it is too springy, it could bounce back up and hit the rack; if it is too hard, the case and/or tail, nose, or other structure could be damaged by repeated ejections. Sandia's Aircraft Compatibility Department has an archive file "CSRL/SRAM Ejection Tests" for pictures and sketches of a cardboard box technique for capturing the SRAM as it was repeatedly ejected at Scot Incorporated, and at Boeing, Wichita. Sandia archive file "B-1 Static Ejection Tests" also shows a slightly different method for capturing bombs during ejection tests. These two files will also show some instrumentation techniques for gathering pitch and velocity data. High-speed video cameras could also simplify the documentation of the events.

5.5 Electromagnetic (EM) test and analysis (RA: SM, SA: SNL/Contractors/AFNWC). Modern military aircraft are exposed to a very complex EM environment. Onboard sources include Ultra High Frequency (UHF), Very High Frequency (VHF), and Extremely High Frequency (EHF) communication transmitters; radars; and electronic warfare equipment such as Electronic Counter-Measures (ECM) pods. The aircraft may also be exposed to significant EM fields due to external sources such as radars and communication transmitters both on the ground and on nearby aircraft. In general, nuclear weapons are designed to be safe and reliable when exposed to a set of EM environments that are specified in the Stockpile-to-Target Sequence (STS). These environments include the EM exposure that a nuclear weapon will see when carried on the aircraft specified in the MCs. When a new aircraft is designed that has a requirement to carry nuclear weapons, the EM environments at weapon carriage points must be determined to verify that they are below the STS limits. Aircraft with an established nuclear weapon capability may have new equipment added to its store list and this new equipment may change the expected EM environments at nuclear weapon carriage locations. Again, the EM exposure that the weapons will see must be determined. Predictions of the electromagnetic fields expected at nuclear weapon locations due to radiators on new aircraft or new radiators on existing aircraft are usually generated by the DoD (by either a contractor or by a DoD engineering laboratory) based on radiator design parameters such as the peak and average power, antenna gain, and physical proximity to weapon locations. SNL engineers may also perform some worst case calculations based on these same parameters. If the field strength predictions are remotely close to the weapon design limits, a test will be conducted on the aircraft to determine the actual field strength.

The Sandia Weapons Systems Engineering Departments are responsible for assuring the weapon will work when exposed to the EM environments specified in the MCs and STS. They conduct tests and analyses, via support from Sandia's Electrical Science and Experiments Department, to assure a capability exists. This activity typically occurs during weapon development, resulting in this task being evaluated far in advance of any other ACCD or CC task. The EM fields need to be reexamined whenever a new aircraft enters the stockpile or whenever changes to an

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existing aircraft occur. The reexamination may be limited to worst-case analysis if the fields are much smaller than the design limits. If the analysis shows that the field may have the same magnitude as the design limits, a test may be required. Testing is conducted by the Sandia Aircraft Compatibility Department using the Electro Magnetic Measurements System (EMMS) trailer and the B61 EMMS test units or by the use of the B61 or B83 Electromagnetic Compatibility (EMC) load units. The EMMS uses two B61 shapes, instrumented with current skin probes, a power meter, and a spectrum analyzer, to convert skin current measurements into EM fields. One unit is exclusively used for ground measurements while the other is flight-certified. The EMC load simulators contain the correct electrical loads of the B61 or B83 gravity weapon so that the correct frequency responses can be replicated when conducting EMC tests. The units also interact with the aircraft AMAC system so that they can be placed in various AMAC states. The respective pullout cables are modified with a current probe and measurements are taken to obtain the direct current injection during an EMC test. The data measured by EMMS and EMC load simulators will indicate if a problem exists that would impact the weapon's compatibility with the aircraft's radiating equipment. When the tests are completed, Sandia reviews the data along with personnel from the weapon systems engineering and electromagnetic departments, and recommends how the ACCD or CC should be written. Depending upon the test results, the ACCD or CC will either be amended to list the new radiating equipment, or it will specify a limitation with this equipment when the nuclear weapon of concern is present.

5.6 Survivability analyses and test (RA: SM, SA: SNL/Contractors/AFNWC). Nuclear survivability requirements are included in the STS for nuclear weapons. Nuclear weapons and their delivery aircraft are designed to be safe and reliable when exposed to the set of nuclear effects environments specified in the STS. Support equipment may experience nuclear effects environments on the ground, which is one phase in the STS. If support equipment is not survivable, nuclear weapons cannot be employed in their prescribed manner. Nuclear environments are typically specified for different operational employment phases of the weapon's life cycle. These environments include thermal, overpressure, gust, and ionizing radiation exposure that a bomb will see when on the ground and when carried on the aircraft. When a new aircraft has a requirement to carry nuclear weapons, nuclear environments must be analyzed in accordance with the aircraft's operational processes and procedures to verify that nuclear effects environments are below the weapon's STS limits. Aircraft with established nuclear weapon capability may have new equipment added to its store list. This new equipment must be assessed to assure that it meets or exceeds survivability levels. Predictions of the nuclear effects environments expected on new or existing aircraft are usually generated by the DoD (by either a contractor or by the AFNWC) based on parameters such as the number, types, yields, height of bursts (HOBs), and Circular Error Probable (CEPs) of weapons expected to detonate near the aircraft. SNL engineers may also perform some worst case calculations based on these same parameters. If nuclear effects environment predictions are more than 25% of the inherent survivability design limits of nuclear compatible equipment, testing of the equipment should be conducted to determine the design has margin.

5.7 Mechanical fit tests (RA: SM, SA: SNL/Contractors/AFNWC). Mechanical fit tests are performed to verify the information provided in the MICDs that was derived from a variety of sources such as actual measurements or 3D computer models. SNL supplied Compatibility Test Units (CTUs) and/or DoD provided Cruise Missiles are loaded on operational aircraft and measurements are made of the spacing and dimensions that are critical to nuclear weapon/ aircraft

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compatibility. These dimensions are compared with those given in the MICDs. The mechanical fit tests are usually performed at the same time as the five-aircraft AMAC electrical interface tests to get a measure of variability across the five aircraft. However preliminary mechanical fit tests may be performed early in the compatibility certification process, and typically during the Preliminary AMAC Test, to uncover incompatibilities in time for them to be corrected prior to the production five aircraft AMAC test.

The task of doing the fit checks to verify the accuracy of the drawings can be completed using CTUs (or other test assets with war reserve outer mold lines), Cruise Missiles, and the best available aircraft hardware. Although the support and/or loading equipment are not part of the MICDs, it is necessary to observe where and how this equipment comes in contact with the weapons and report any concerns to the users.

Often it is useful to take an early draft of the drawings to the field and "piggy-back" on some other activity in order to determine voids or incomplete data. However, before drawings are signed off, a full-scale fit-check has to be accomplished. For internal carry, doors open for release and doors-closed clearances must be verified. Landing gear retractions are sometimes required to validate clearances on external carriage, along with speed brakes, ailerons, flaps, and personnel access to the bomb rack and pullout cable connections. Taking still photos or video of the mechanical fit is a good way of documenting the test for future reference. The agencies that will sign the MICD should be present at the final fit test event.

5.8 Vibration flight tests and analysis (RA: SM, SA: Contractors/AFNWC/SNL). The purpose of the VFA tests and analyses is to determine the vibration environment imposed on components inside the nuclear weapons when carried on the aircraft. The aircraft with the VFA test unit loaded in the configurations of concern, is flown at various Mach number-altitude points within the desired flight envelope that, by analysis, have been determined to provide the worst-case vibration environments to the weapon. The vibration spectra obtained from this flight testing must be lower than the spectra for which the weapon components were originally designed in order for the aircraft to be judged as compatible with a particular nuclear weapon. In some cases, certain flight restrictions are placed on special configurations that do not pass the tests at all points within the flight envelope. The aircraft in some tests are configured in what are considered to be worst-case configurations, and other less severe configurations may be cleared by analysis.

The VFA test units are supplied by SNL. They may use either air-to-ground telemetry or an onboard recorder to acquire data. The VFA test units duplicate the War Reserve (WR) mass properties and will have the real components or mass mockups of the components of concern installed. The components of concern are instrumented with accelerometers. The aircraft are supplied and operated by the DoD; the tests are staged out of a DoD test base and flown over a DoD or DOE/NNSA test range.

One of the challenges with this task is pre-planning all of the necessary activities to get all of the flight test data needed for certification. Weapons are carried externally or internally and the aircraft can be either operational or an early production model. Thermal tests using VFA test units would include ground and flight data to support compatibility of the weapon with MC and STS temperatures. For external carriage, compatibility analyses can be accomplished if the mach/alti-

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tude and time at those conditions can be provided. For internal carriage, the aircraft contractor should provide temperature profiles experienced during ground and flight operations. A good place to start is to obtain and look at the temperature design requirements used for aircraft equipment installed in the weapons bay and/or pylon station. The next step is to have the aircraft contractor provide temperature profiles based on instrumented units used during flight tests. If required, an instrumented unit could be provided by Sandia. Data from Sandia VFA assets is analyzed by Sandia's Program & Test Integration in conjunction with Aircraft Compatibility and Weapon Systems Engineering.

Here are three examples of how VFA data was acquired in the past:

5.8.1 PA-200. Sandia could not use telemetry frequencies in Germany because of interference with their TV channels. Planning began a year ahead of the test with a designated test aircraft that had extra tracks available on an existing on-board tape recorder to record data. Co-axial wiring was added between an external fuselage bomb station and the recorder, including an on-off 28 Volt DC control switch in the cockpit. Sandia range support personnel shipped over a checkout and data playback trailer along with a diesel engine driven 120 Volt AC-60Hz generator. Prior to flight Sandia ran a few minutes of tape and played it back in the trailer to check on correct operation of all channels. After the flight, the stripped-out tape tracks were reviewed and if the data was "good" a duplicate was made and at the end of all the tests, the tapes were hand carried back to Sandia for data reduction.

5.8.2 F-16 with LANTIRN. In 1990 the USAF added LANTIRN to the F-16 aircraft. This required that additional testing be performed to evaluate the change in boundary airflow around the bomb caused by the addition of these new external pods on the forward fuselage of the Block 40/42 F-16C, D aircraft. The USAF made a just-off-the-line production aircraft available for these tests and would not allow any changes or additions to this combat-ready aircraft. In this case, Sandia used the aircraft integrated AMAC system and the bomb CF2438 pullout cable to control the air-to-ground T/M system. The VFA test unit had a small antenna mounted on the bottom of the after body to get the signal to a Sandia ground station for checkout or to a tracking radar dish at Vandenberg AFB, CA, which was linked to a ground station during the flights. The Vandenberg ground station was manned by Sandia personnel who made duplicates of the data flight tapes and reduced the data to compute Power Spectral Densities (PSDs). The F-16C was configured without external fuel tanks, which required tanker support, thus allowing all the high-speed data runs for one bomb configuration to be completed on one flight (SAND 94-2845).

5.8.3 B-2A Internal Carriage on Rotary Launchers. Flight testing began in the spring of 1994 at Edwards AFB. Air Vehicle No. 2 (A/V-2) had been selected to accomplish this task and a dedicated tape recorder was installed in the aft equipment bay. A/V-2 had no requirement for AMAC software in the system to control a nuclear bomb; thus, a 28-Volt DC on-off control switch was installed in the right pilot's station plus hard wiring to a bomb location on the rotary launcher. There was also an indicator in the cockpit that showed the amount of recorder tape available in seconds. The plan was to load the VFA unit on the launcher in the Integrated Maintenance Facility (IMF), check the wiring in the launcher, check the wiring in the aircraft, load the loaded launcher in the aircraft, and check the system by running a few minutes of tape and playing

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it back in a Sandia-provided ground station trailer. This same ground station was used to do on-site data reduction on the data tapes received after each flight.

5.9 Non-Combat Delivery Vehicles tests and analysis (RA: SM, SA: Contractors/AFNWC/SNL). Non-Combat vehicles are tested to verify that the vehicle does not allow environmental inputs (e.g. for example, shock or vibration) into the weapon that might exceed weapon designed component levels. The tests will use representative test assets of sufficient high fidelity to ensure capturing realistic environmental data. The mechanical compatibility test shall also include monitoring and analysis of static and dynamic inputs to the weapon produced and transferred by the Non-Combat Delivery Vehicles. Some common environments per the STS are transportation loads and mate and de-mate loads.

5.10 Weapon separation tests/analysis (RA: SM, SA: Contractors/AFNWC/SNL). The purpose of this work is to assure the safe release of weapons from aircraft throughout the intended release envelope as described in the Concept of Operations (CONOPs). The test units used for weapon separation drops are generally Bomb Dummy Units (BDUs) supplied by the DoD, but may be weapon development units supplied by SNL. The data acquired in the weapon separation tests is usually in the form of video taken from the release aircraft or from a chase aircraft. Computer analysis of the video to determine trajectory, orientation, and other important parameters is performed by the appropriate agency. This data is used by the certification authority to verify safe separation, bomb fall lines, clearance with adjacent structure/stores, separation model accuracy, and other information. The video and computer-generated data are supplied to SNL for their parallel analysis. SNL engineers use this information to verify that the nuclear weapon is released from the aircraft within the proper aerodynamic parameters. In a retarded drop, the weapon must not pitch or yaw beyond certain limits so that the parachute will operate correctly. Tests are also conducted at the extremes of the flight envelope to verify repeatable ballistics for freefall units or proper decelerations to operate weapon Environmental Sensing Devices.

It is important to note that a "safe" separation from an aircraft point of view means that the bomb will not hit the aircraft; however, a safe separation from an aircraft viewpoint may not be satisfactory from a weapon standpoint. Examples of this unsatisfactory performance include trajectories that do not allow the weapon to function properly or trajectories that produce excessive side load on the bomb when a parachute is deployed. Thus, it might be a safe separation, but from a weapon standpoint it might not be satisfactory.

Sandia engineers found that BDU-38s produce poor separation results because they are not good physical representatives of the B61 mods currently in the inventory (wrong weight, center of gravity, and moments of inertia). Stores that closely match the WR physical characteristics should be used for maximum commonality.

As an example of how things could happen, years ago a B28 test unit was released from the weapons bay of a B-52. At about one bomb length below the aircraft, the bomb had rotated 90 degrees to a nose-down attitude. The velocity-sensing devices were blanked out so arming could not occur. In addition, the marginally unstable bomb continued to tumble producing an unacceptable trajectory - it would never hit a designated target. The contractor who was demonstrating releases off a rotary launcher said it was "safe." Sandia said it was unsatisfactory. After reposi-

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tioning the bomb on the launcher and by changing the bomb rack ejector forces, subsequent drops were safe and satisfactory.

Another "rule" for satisfactory separations where the parachute is deployed is that the pull angle of the deployed parachute should be less than 12° with respect to the bomb centerline, otherwise, parachute shroud lines could be cut or damaged on the bomb tail can or the tail can could be damaged by excessive side loads generated by the off- angle pull of the parachute shroud lines.

The parachute deploy delay times to be used operationally should be demonstrated and used during separation tests. Factors to be considered are: a) to determine that the nose-down pitch angle and the distance below the aircraft are adequate to prevent parachute deployment devices from hitting the aircraft, be aware of deflected aircraft control surfaces; and b) to be aware that for some low-level retarded releases there could be a release altitude coupled with an ejection velocity where the bomb will hit the ground before the bomb has had a chance to trajectory arm. The options then are to trade off delivery altitudes, ejection velocities, and parachute deploy delay times until the user is happy with the solution.

At the beginning of a compatibility program involving a new aircraft to employ a stockpiled bomb, this question is frequently asked of the Sandia engineer: "How many drops do you require?" The answer is usually "four," as follows: One retarded at low- altitude, minimum parachute q; one retarded at low-altitude, maximum aircraft or parachute q; one high-altitude free-fall at close to minimum aircraft q; and one high- altitude free-fall at maximum aircraft q. The DoD will usually provide more than four drops. The aircraft contractor aero and test teams will select their first drop to be at a very benign condition, usually a mid-envelope Mach/altitude test point and then work their way around the flight envelope as flight clearances are provided by the flight safety test teams. If their wind tunnel predictions agree with actual drops, then the tests will follow a logical build-up program. Computational Fluid Dynamics modeling can also be used if the data is available.

5.10.1 System 2 Umbilical Testing (RA: SNL, SA: SM). System 2 (S2) adapters provide an electrical connection between the weapon delivery system (through the umbilical cable) and the weapon control electronics. Umbilical cables and connectors for System 1 (S1) are furnished by SNL and all quality assurance criteria are performed in-house, while S2 umbilical cables are provided by the aircraft platform program offices. This change in responsibility moves the bomb-to-aircraft interface from the aircraft side of the umbilical connector down to the weapon side of the umbilical connector. As such, Sandia National Laboratories must test the interaction of the S2 umbilical and the bomb during captive carry and ejection in order to ensure proper functionality between the aircraft and the weapon system. Specifically these tests must evaluate satisfactory timing of the aircraft and weapon messages and signals.

5.11 Develop Mechanical Interface Control Drawing (MICD) (RA: SM, SA: Contractor/SNL/AFNWC). The MICD document depicts the physical configuration of an aircraft loaded with nuclear weapons per the requirements of Data Item Description (DID) DI-NUOR-81407. 3D models are preferred and are generally produced by a DoD contractor or by the USAF. The MICD is signed by the originator, the System Program Office, the AFNWC, and SNL. The signatures are an indication that the drawing or definition is complete and that it fully depicts the aircraft and the

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installation of nuclear weapons. Revisions to the MICD are submitted for review and approval. All signatories must approve the revised MICD.

The MICD consists of at least the following:

a. Signature block for responsible agency representatives.

b. The aircraft nuclear weapon carriage configuration.

c. Aircraft Alternate Mission Equipment (AME) configuration with nuclear weapon installed.

d. Aircraft weapons bay configuration with nuclear weapons installed.

e. Ejector rack, pullout cables, and other details.

f. References to drawings to include the aircraft, the ejector rack, the appropriate nuclear weapons, and other items such as ECM pods, fuel tanks, and missiles.

The MICD is one of the documents that will be used and maintained throughout the life of the aircraft/nuclear weapon system. For a specific aircraft, there should be a separate drawing or definition for each nuclear weapon required on the aircraft so that if a nuclear weapon is deleted or added, then that MICD is deleted or a new one added and the documentation will track the capability.

Typically the first sheet of a hard copy MICD or Computer Aided Design (CAD) MICD should list aircraft version/configuration/equipment and contain all the notes and references, a signature block, and if possible, a 3-view of the aircraft. The next sheet should show "big picture" views of pylons, launchers, etc., and then follow-on sheets should show smaller views with additional details. The notes should include the parachute and spin rocket deployment time selected for this application, the cartridge and orifice callouts, and the required pullout cable for each nuclear weapon.

The MICD must be completed before the mechanical fit tests accomplished during the five aircraft AMAC tests so the drawing can be checked for accuracy.

5.12 Develop AMAC Preliminary Design Report (PDR) and Final Design Approval Report (FDAR) (RA: SM, SA: Contractor/SNL/AFNWC). To document the AMAC design, the System 1 (SYS 1001) and 2 (SYS 2001) specifications require that the AMAC designer submit to their aircraft System Program Office a PDR and a FDAR, per the requirements of DI-NUOR-81408. These reports are then submitted to the AFNWC and SNL. The AFNWC and Sandia agencies then review the documents and provide recommended changes and eventual approval of the documents back to the aircraft SPO.

In the early stages of a new aircraft design, a PDR is produced and reviewed in order to aid the designer in producing a certifiable AMAC design and supporting the preliminary AMAC test. The PDR consists of a report of hardware and software information which must describe and

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define the design concept to the extent that the components, as well as the operational capability of the system, are clearly understood. The PDR is prepared and presented by the AMAC designer to AFNWC and SNL at a time sufficiently in advance of final design submittal, usually in the Engineering & Manufacturing Development Phase, as to allow for design changes that can be implemented without impacting the program's schedule. Any changes occurring after PDR must be concurred by all concerned prior to inclusion in the FDAR.

The FDAR is a document required by the AMAC specifications that documents the AMAC design and ensures the AMAC system meets the AMAC specifications. It contains an electrical analysis of the AMAC system, including power supply characteristics, switching logic, monitoring circuitry logic, individual component characteristics and digital or analog control logic. In the case of an aircraft with an integrated AMAC, such as the F-15 (SAND 92-0828), F-16, and B-2, it also contains a description of the cockpit displays and software used to control the weapon interface. An FDAR shows a comparison of conservative and worst case design results obtained from analysis versus AMAC specification requirements. The FDAR is prepared by the designer of the AMAC system. The FDAR must cover all aspects of the AMAC design since this document will be used throughout the life of the aircraft. It should not be approved until all required information, as specified in the AMAC specification, are included. Once the FDAR is approved, a requirement in the DID DI-NUOR-81408 states that the FDAR must be kept updated so that it represents what is currently in the inventory. Information is added to the FDAR whenever changes are made to the AMAC hardware or software. Funding for this kind of update is provided by the aircraft SPO's. The FDAR must be submitted for approval a minimum of 60 days before the five aircraft AMAC test is conducted.

The AMAC Design Approval Procedures defined in DID DI-NUOR-81408, states the requirements for the kind of information that must be included in an FDAR. Occasionally the AMAC designer misinterprets the specification requirements and does not include all of the analysis required. For example, the System 1 loads represent the worst-case condition. The FDAR analysis of these loads on each pin is to be calculated at the worst-case temperature, typically the highest temperature to which the AMAC system can be exposed. Sometimes the designer only calculates at ambient temperature.

The DID DI-NUOR-81410 for the EICD states that the FDAR should be included in the EICD, but the FDAR can be delivered separately from the EICD. Once the FDAR is approved, any future changes to hardware and/or software requires the FDAR to be amended with the new information and analysis, and resubmitted to the AFNWC and SNL for review and approval. The FDAR must be approved before any AMAC testing is conducted on the updated AMAC system.

The System 2 AMAC specification exists between the nuclear weapon and a single MIL-STD-1760 umbilical cable. The specification defines message format requirements needed to provide monitor and control of a nuclear weapon for the serial digital data interface, the electrical power provisions, the address line requirements, interface discrete requirements, and the multiplex data format between a nuclear weapon and a carrier aircraft for AMAC functions. Other messages are defined in the Platform Store Interface Control Document (PSICD). All stated requirements are mandatory for the carriage and operation of a nuclear weapon. The System 2 requirements, in addition to requirements levied by the PSICD, supersede the Military Standards

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where conflicts of requirements appear. Nuclear certification AMAC testing by the AFNWC and SNL, verifies not only the requirements of System 2 spec, but also that of all documentation requirements that makes up the All-Up-Round definition. The FDAR and EICD are the prime source documents used in the preparation of AMAC testing.

5.13 Develop EICD (RA: SM, SA: Contractor/SNL/AFNWC). To document the AMAC design, the System 1 and 2 specifications require that the AMAC designer submit to their aircraft System Program Office an EICD per the requirements of DID DI-NUOR-81410. The EICD is a signature-controlled drawing set detailing the aircraft/nuclear interface and associated electrical systems. The EICD must contain information extracted from contractor documentation in order to complete a composite drawing reflecting the basic electrical interface features. It is provided to the AFNWC and SNL in the same time frame as the FDAR. Revisions to the EICD are submitted to the AFNWC and SNL for review and approval. The EICD is signed by the originator, the System Program Office, the AFNWC, and SNL. The signatures indicate that the signatories agree that none will make changes to the EICD without informing the others, all agencies approve the design, and agrees on the accuracy of the package's contents. The revised EICD must be approved by all signatories before it is accepted by the appropriate SPO.

The EICD consists of at least the following:

a. Signature Block for responsible agency representative.

b. System Block Diagram of the aircraft AMAC and release electrical systems.

c. Aircraft AMAC system schematic.

d. Aircraft release system schematic.

e. Aircraft AMAC and release component descriptions.

f. Aircraft AMAC and release power source descriptions.

g. List of drawings that define the nuclear weapons, ancillary test equipment, and training devices appropriate to the aircraft.

h. FDAR (provided separately).

The EICD (also known as a Weapon System Electrical Compatibility Control Drawing on the B-52 and Electrical System Description Drawing on the B-2) is a very important and useful document. It is used as a requirement for releasing an ACCD and NCCS, as a way of understanding how the AMAC and release system works, as a tool for troubleshooting when problems in the field occur, and as an aid in the design of future weapon systems. It is the only electrical drawing comprised of the entire aircraft designer's drawings used for building the complete aircraft system. Without this single drawing package, an enormous amount of documentation would have to be researched to piece the AMAC and release systems together as it exists from the aircraft engine power source to these interfaces. The EICD is structured this way to show the reader how the sys-

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tem flows (from left to right in a logical manner) and all the components in the aircraft that make up the power source, AMAC, and release systems.

The FDAR is included in the EICD package because it contains all of the analysis necessary to prove whether or not the AMAC system meets the AMAC specification. The list of nuclear weapons, ancillary test equipment, and training devices that could be connected to the AMAC system, is included in the EICD to define what equipment will be used throughout the life of the aircraft.

The EICD, like the MICD, is updated throughout the life of the aircraft. Sometimes it is difficult to get information added to a signed off EICD or MICD because funds may not be available from the aircraft SPO to update the drawings. Signoff will occur every time there is a change to these drawings so that they will represent the current configuration of the AMAC and aircraft release systems. The Electrical System Description Drawing on the B-2 is an excellent example of what an EICD package should look like. When the AFNWC and SNL engineers and the aircraft designer are developing an EICD, it would be worthwhile for them to review this drawing package, and for the AFNWC engineer to provide a copy of the B-2 Electrical System Description Drawing (ESDD) to the aircraft designer for reference. This will make the engineer's review of the EICD much easier when it is time to approve the drawing.

5.14 Preliminary weapon/aircraft electrical interface tests (RA: SM, SA: SNL/Contractors/AFMWC). A preliminary AMAC test on an aircraft should be performed as early in the program as possible. The test may be run on laboratory hardware before full aircraft hardware is available or they may be run on test aircraft that are being used to evaluate the new design, or both. They can be performed at a contractor's facility in the case of lab tests, or at a military facility that has the appropriate aircraft available. Preliminary aircraft/weapon electrical interface tests are conducted after a Preliminary Design Report (PDR) is submitted to the AFNWC and Sandia. Conducting a preliminary AMAC test in the aircraft integration lab when hardware and software become available, and before going to the aircraft, is strongly encouraged. A preliminary AMAC test on an aircraft provides a better representation of the final five aircraft AMAC tests, than a test of aircraft hardware and software in a laboratory. A preliminary AMAC test on an aircraft also helps the AFNWC and SNL engineer's mechanical counterpart conduct preliminary mechanical fit tests using CM's or CTU-1's or -2's mounted on the aircraft. When the preliminary AMAC test is conducted on an aircraft, the test plan should be done in collaboration with the AMAC designer, Sandia, and AFNWC engineers. This helps determine if the test plan will be adequate for the five aircraft AMAC test, and it will also provide data for comparison. AFNWC and SNL will run the test with its AMAC test equipment, and the results are shared with the engineering community that has responsibility for the design of the aircraft nuclear system.

Preliminary aircraft/weapon electrical interface tests are not a requirement for the release of an ACCD or CC, but they are conducted as part of a nuclear compatibility certification program required for a new aircraft or for an aircraft undergoing significant modifications. The PDR supports the preliminary AMAC test. The five-aircraft AMAC tests, described in the next section, are conducted for the purpose of verifying that operational aircraft meet their nuclear compatibility certification requirements. The five-aircraft AMAC tests are accomplished on operational aircraft and it is expected that they will pass these tests. If they do not, it may mean that a retrofit

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needs to be performed on the aircraft. A retrofit effort will probably result in a delay in the initial nuclear weapon operating capability for the aircraft since the five-aircraft AMAC test typically occurs after the aircraft have entered the production phase.

5.15 Weapon/aircraft interface and AMAC electrical tests (RA: AFNWC, SA: SNL/Contractors). These tests are known as the five aircraft AMAC tests, and are performed jointly by AFNWC and SNL on operational aircraft with operational crews, at an operational base. The Aircraft Contractor is also present in support of this series of tests. The purpose of the AMAC electrical interface testing is to ascertain compliance of the aircraft AMAC system with the required AMAC specification, the MICD, and EICD, and to establish that the aircraft is indeed electrically and functionally compatible with the required set of nuclear weapons. The AMAC tests are typically some of the last tests to be run in a nuclear certification program. The tests are performed on five production aircraft that have been prepared according to USAF Technical Orders (TOs). The number five was chosen, based on experience, to be large enough to detect variability between aircraft yet small enough to make the test program tractable. The AMAC testing is divided into two parts: ground and air. The ground tests are, in turn, divided into AMAC specification compliance tests, MICD compliance verification, EICD verification, and functional tests using CTUs. The air tests, which may require more than one flight test, consist of functional tests defined by SNL to verify end to end weapon system compatibility. The AMAC System 1 and 2 specification compliance ground tests, conducted by the AFNWC with the Special Weapons Interface Tester (SWIFT) checks the ability of the AMAC system to provide the electrical and digital data interface required by the appropriate AMAC specification, and other supporting documentation like Weapon Specific Addendums and Platform Store Interface Control Documents. The AFNWC has also published, through the AMAC POM, a System 1 and System 2 Test Requirements Documents that define what is tested at these interfaces. After the SWIFT tests are accomplished, Sandia conducts functional tests with Compatibility Test Units (CTUs) used to evaluate the ability of the aircraft AMAC system to properly safe, operate the Permissive Action Link (PAL), and pre-arm nuclear weapon components.

AMAC tests are conducted utilizing the following test assets, tools, and methodologies:

5.15.1 Compatibility Test Units. CTUs, which are virtually identical to WR nuclear gravity weapons (except for a dummy warhead), are electrically connected to the aircraft AMAC through a pullout cable to monitor and control the nuclear weapon. The aircraft AMAC system is then used to safe and pre-arm the CTU (including PAL, USG, and CD operations), and the electrical signals transmitted between the aircraft and the nuclear weapons are recorded. For the ground electrical functional interface tests with CM's and/or CTU's, the AMAC may be powered by either aircraft power or by auxiliary power carts, since functionality and sequencing are the important parameters. SNL has three battery-powered data acquisition system testers for measuring interface signals with CTUs: a SE3331 for legacy weapons and two Functional AMAC System Testers (FASTER) for the B61-12; the 1E5223 flight recorder, and the 1E5472 ground recorder. The SE3331 and 1E5223 are mounted in specially designed CTUs used to record data transmitted between the aircraft and the nuclear weapon. The data is read out and analyzed while on the ground and after the flight is completed with SNL provided ground data and flight data acquisition recorders. Flight data acquisition recorders are located inside the CTU and data is downloaded on the ground after the flight is completed. One to five in-flight functional interface

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tests are accomplished for each nuclear weapon type to be evaluated. SNL will determine the number of flight tests needed to accomplish certification and will coordinate the test with AFNWC. CTUs are built specifically for Sandia's Aircraft Compatibility Department for AMAC testing purposes. There are five types of gravity weapon CTUs:

5.15.1.1 CTU-1. A classified flight-certified unit that is electrically and mechanically equal to a real nuclear weapon except it does not have the physics package. It uses ballast in place of the physics package so that it will represent the same weight and center of gravity (Cg).

5.15.1.2 CTU-2. Identical to a CTU-1 except it has less sensitive components. It may or may not have classified material and is used in AMAC tests that have a multi-station capability.

5.15.1.3 CTU-3. Classified like a CTU-1, but it does not have a nose or tail assembly and is not flight-certified. It is used in situations where space is at a premium, like in a lab environment, but where classified material is not a problem. There are no CTU-3's for the B61-12 as the Tail Kit Assembly is an integral part of the weapon design to the AMAC interface.

5.15.1.4 CTU-4. A CTU-4 is an unclassified suitcase version of a CTU-2. It is used where portability is the key factor and is not flight-certified.

For the Air Launched Cruise Missile (ALCM) there is a classified and unclassified version of the W80-1 CTUs. The classified version contains a W80-1 firing set that does not contain explosives or hazardous material. It contains all of the WR electrical safety and use control components and is housed in a suitcase. The W80-1 unclassified unit is also housed in a suitcase. It contains the USG safety component but the PAL and CD loads are simulated. When a complete load out configuration is required for testing 20 ALCMs on B-52H aircraft, Sandia has small unclassified W80-1 Type 2D Prearm Load Simulators (PLS). The PLS is unclassified and is used in place of a W80 warhead on ALCMs to provide warhead circuit loads for prelaunch operations while recording electrical interface signals and environmental data, including shock, vibration, temperature, humidity, and barometric pressure. The PLS is flight certified when installed on a warhead surrogate shape. These units allow test engineers to conduct a 20 missile prearm test on the B-52H aircraft (SAND 91-2664). Its small size (1-1/2 x 3 x 8 in) allows it to be installed on USAF training/ferry payloads to acquire flight or ground tests data. The W80-1 TYPE 2D PLS replaces the earlier W80-1 PLS, which did not include data acquisition capability. They simulate the electrical interface portion of the warhead and its small size allows it to be used with USAF training/ferry payloads when flight testing is required.

The AMAC designer should always thoroughly test the AMAC design at the AMAC interface using the requirements specified in the System 1 or 2 specifications and supporting documentation (SYS 1300, SYS 2300, and SAND 97-3147). The SNL Aircraft Compatibility Department CTUs and PLSs can be loaned for a short period of time so that the AMAC designer can use WR-equivalent hardware in their functionality tests.

AMAC tests are very time consuming because it will probably be the first time everyone involved in such a test on the aircraft is present. A maximum of one week for the preliminary test and one month for the five aircraft tests should be planned. These timeframes take into consider-

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ation testing a multiple station aircraft for the SWIFT tests, a number of different multiple station weapon configurations for the ground tests, and multiple station flight tests. No other maintenance or test operations are allowed on the aircraft while the test is being conducted. This will assure that the engineer will not be preempted at any time.

5.15.2 Verification of AMAC Cockpit displays. FDAR and EICD verification of cockpit displays are accomplished during ground AMAC tests by AFNWC and Sandia. This can be accomplished by test procedures and the recording of AMAC displays during the joint AFNWC and SNL tests.

5.15.3 AC and DC Load Banks. Sometimes an AC or DC load bank is required during the AMAC tests. The way the aircraft's power source to the AMAC interface is designed will determine if an AC, DC, or perhaps no load bank is needed. This is done to load down the aircraft's power supply so that it represents the kind of loading that is either present during a nuclear delivery mission, or a load that represents the maximum capability that could be imposed to the aircraft's power source. The aircraft designer is required to analyze the worst-case and nuclear weapon delivery load conditions in the FDAR; this established value is used as a setting for the load bank(s). The AMAC designer can provide the AFNWC and Sandia engineers with these load conditions early enough in the program so that the engineers can use this information for both the preliminary and final AMAC tests. The load bank is used with aircraft engine power, not with a ground Auxiliary Power Unit (APU). Typically, a ground APU cannot handle the added load from the load bank(s), which will result in the aircraft not meeting the AMAC specification voltage requirements. There may be aircraft that have an onboard APU capable of handling the added load. In these cases, turning on one or more engines are not necessary to conduct the AMAC tests. An APU source of power is acceptable if a change in the aircraft's AMAC software is the only test being conducted. Typically, this is done in follow-on AMAC tests after the five aircraft AMAC tests have been completed. There may be aircraft (i.e., the B-2 and PA-200) that have such stable power supplies even at maximum loading that no problem exists meeting the requirements of the AMAC specification. With this type of aircraft, a load bank may not be necessary in follow-on AMAC tests (SAND99-0400 and SAND92-1527).

5.15.4 Simulated Flight Test. Since the USAF are responsible for assuring the release system is working properly, they may ask the Sandia Aircraft Compatibility Department to conduct a simulated flight test on the ground to verify everything works properly when any type of release command is generated. A release command can be generated manually or automatically, and in the case of an emergency, by ground or air jettison commands. Running engines may be required when simulating flight on the ground to gather data not possible in the air. For example, because CTUs are not released (i.e., drop tested) from the aircraft, what occurs at the AMAC interface when an air jettison command is generated would need to be measured.

5.15.5 Flight Test. Sandia provides step-by-step procedures that the crew member will use during the actual flight test. These procedures must be precisely written so that there is no operator confusion. (It is best to avoid a mission being scrubbed because of a problem with the test plan.) The AFNWC and Sandia engineers may have the opportunity to review the procedures with the crew member before the flight; however, this may not occur if it interferes with the

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required crew rest period before the flight test. An accurate Flight Test Plan is beneficial in two ways:

It makes crew debrief after the flight much easier because the procedures contain exactly what kind of indications the operator should see when going through the various AMAC operations. This is necessary because not all of the CTUs flown on the flight test may have the flight data recorder installed to record what occurred at the interface.

On stations that do have a flight data recorder configured CTU, it helps on the data reduction to determine if the aircraft and bomb are responding according to the commands given by the operator in flight.

5.15.6 Conducting AMAC Tests. Both the AFNWC and SNL create test plans and publish test results of AMAC tests. Test results are reported to outside agencies, document what conclusions were drawn, and what capabilities were granted based upon the test data and the approved FDAR and EICDs. If the aircraft fails to meet any part of the AMAC specification or Platform Store Interface Control Document, the AFNWC and SNL engineers must inform the AMAC designer and the government agency in charge of the program so that the problem can be remedied. First, it must be determined if the AMAC design can be fixed to meet the requirements. If this is not possible because of program cost considerations, or for any other reason, the Sandia engineer will probably be asked to verify that the weapon(s) carried by the aircraft will work properly in the area that is out of specification limits. Sandia's Weapon Departments can assist with this kind of analysis. The AFNWC and Sandia Aircraft Compatibility Department also has data in their files that define past problem areas. This documentation can be found in the various aircraft and weapon archive files. Depending on the complexity of the problem, a limited capability or no capability at all may be stated in the ACCD and/or CC(s).

AMAC tests are also performed on a new weapon or a modification to an existing weapon. For these situations, an AMAC test is conducted using CTU's and the SWIFT for the ground tests and the CTU's and the flight data recorder for the flight test. Upon successful completion of these tests, and if necessary the Full Weapon System Drop Test, the weapon is added to the ACCD and/or CC. There is also the situation where the Sandia Weapon Departments determines that a modification to an existing weapon is so minor that it is added to the ACCD or CC without any AMAC testing accomplished. Attachment 4 provides a checklist to aid the AFNWC and Sandia engineers to determine when and what kind of AMAC electrical tests are required based upon the type of changes made to the aircraft platform. Periodically, after the five aircraft AMAC tests, the aircraft may undergo software and/or hardware changes that may affect the AMAC system. These changes are documented in updates to the FDAR. The AFNWC and Sandia engineers determine if the changes are significant enough to warrant follow-on AMAC testing to confirm the analysis in the FDAR. If a change is minor and no AMAC testing is necessary, a capability can be granted by adding it to the appropriate NCCS, ACCD, and CC. When testing is necessary, the Sandia engineer will work with the AFNWC to create a ground and/or air test plan(s) that defines what needs to be accomplished.

5.15.7 Comprehensive AMAC Tests. Sandia requires testing every 5 to 10 years on at least two aircraft. This AMAC testing is different from the surveillance tests conducted annually

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by the AFNWC. During surveillance testing, only the aircraft side of the interface is tested with ground power, SNL test assets are not used for surveillance testing as the AFNWC is only ascertaining whether or not there is any aircraft AMAC system degradation. For the comprehensive AMAC tests, aircraft engines are running and AC or DC Load Banks may be used, since interface voltage levels under load are critical. The 5 to 10 year timespan is based on what the Sandia Aircraft Compatibility Department has done in the past, and it fits in well with updates made to the aircraft. Both ground and flight tests are accomplished using the SWIFT, FASTER, and flight data recorder testers. The purpose of these tests is to verify that no degradation to the power source or AMAC system has occurred that would cause the aircraft to fail to meet the AMAC specification requirements. Another purpose of these tests is to provide and maintain a baseline for other, more minor AMAC tests. A failure due to degradation may jeopardize the aircraft's capability as stated in the ACCD and CC(s) and possibly in the MNCL.

Besides conducting AMAC electrical tests on aircraft, there are also various AMAC test facilities available to conduct tests and each have a very specific purpose and level of fidelity. AMAC Laboratories at the AFNWC and SNL are used to check out the SWIFT, FASTER, and Flight Data Recorder testers prior to shipping equipment to a base. System Integration Labs (SILs) are located at various aircraft contractor facilities and Air Force Bases responsible for developing or maintaining aircraft weapon systems. SILs are useful facilities for checking out development software to determine if the software integration with AMAC hardware will function properly and meet AMAC specifications and Platform Store Interface Control Documents. Some SILs use nuclear weapon simulators to provide simulation of a nuclear weapon. On occasion they have also requested the loan of Sandia CTUs to verify proper operation of weapon components. SILs are not used in place of aircraft for certification or surveillance testing. They do not represent the entire aircraft as far as electrical loads, power supplies, or total number of nuclear weapon store mechanical or electrical configurations.

5.15.8 AMAC Data Reduction. Reducing the vast amount of data taken during a ground and air AMAC test can be very time consuming. The SWIFT test results are designed to show where the data falls within the pass/fail criteria when compared to the requirements. Flight test data reduction has to be compared to the AMAC command and known weapon responses to that command. Like in the SWIFT, there are analysis programs for PAL, USG, and CD in the flight data recorder that flags when the specification requirements are not being met.

5.16 Full Weapon System Demonstration/Drop Test (RA: SM, SA: SNL/AFNWC). The purpose of this test is to exercise the whole weapon system from beginning to end. SNL and NNSA provide a nuclear weapon test unit for this test that, as nearly as possible, duplicates the features of a real WR nuclear weapon. The test units are transferred to the USAF who then handles the weapon as if it were real. These units are very similar to WR assets converted to Joint Test Assembly (JTA) units; except they are not "scored" on the accuracy of the drop and they are not taken out of the stockpile. It is necessary to use operational aircraft, maintenance crews, weapon loading crews, flight crews, and Technical Orders. The aircrews are given the task of delivering the test unit to a target. The target is typically an instrumented test range, such as Tonopah Test Range, where the delivery parameters of the weapon can be measured as the weapon separates from the aircraft and falls to the target. This is the only test that evaluates the entire system and it is very important, because it is the final task for the requirements to release an

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ACCD and CC. It is the most realistic test of the STS and is done to ensure everything (aircraft and weapon) works correctly. It is required because all other ACCD and CC tasks focus on a specific part of the overall aircraft/weapon system.

For example, an AMAC test checks only that portion of the aircraft that safes and prearms the weapon, because it is an AMAC specification area of concern. Other tasks described in this Handbook are concerned with the following during this test: Aerodynamic loading on the weapon because of various flight conditions, individual weapon components and the assembly as a whole reliably operating in the environments to which they are exposed, and weapon interfacing properly with the aircraft.

The successful results of this task, combined with all of the other tasks described in this Handbook will result in the publication of the ACCD and/or CC by Sandia's weapon systems engineering departments, and the publication of the NCCS by the AFNWC.

5.17 Joint Test Assembly (JTA). The JTA Program is managed by NNSA with Sandia and the DoD as major contributors. The DoD treats this test as a "scored" mission, meaning the crew is evaluated on their proficiency to deliver a nuclear weapon. Sandia's Stockpile Evaluation Department is directly responsible for the JTA test. The Sandia Aircraft Compatibility Department helps the Stockpile Evaluation Department understand how the aircraft AMAC systems work and what is unique about each one of the systems. If a failure in the test occurs, both Departments analyze the failure to determine if it was caused by the aircraft, the weapon, or the crew.

In years past, it was possible to have a JTA mission on an aircraft whose AMAC software had yet to be included in the FDAR, ACCD, or tested by the AFNWC and Sandia. With the advent of the MNCL, a confirmation is now made by the AFNWC and Sandia, that the operational software, planned to be used for a JTA mission, is listed in the MNCL as officially approved software to be used with nuclear weapons.

On a new aircraft, pulling a WR unit from the stockpile may not fit into the aircraft's nuclear certification schedule. It might be a better plan to have the weapon project groups provide a bomb or warhead asset for the Full Weapon System Drop Test so that other JTA activities can proceed on their own schedule.

5.18 Publish Statement of Compatibility (SOC), ACCD, and CC (RA: NNSA/SNL). Publishing the SOC and releasing the ACCD and/or CC are the final certification tasks performed by SNL. The SOC is an SNL/NNSA letter to the AFNWC documenting the nuclear weapon system compatibility with a specific weapon. The SOC is issued by SNL ensuring the ACCD and CC documents will be updated. The ACCD is a document that is referenced by the Major Assembly Release (MAR) for each nuclear gravity weapon, and addresses compatibility with the carrier aircraft designated by the MC or STS. Each ACCD contains a listing of the aircraft capable of carrying the nuclear weapon, and documents any conditions and restrictions for loading, carriage, and delivery. The ACCD documents that SNL and NNSA are satisfied that carriage and release of the nuclear weapon on the aircraft will not affect the assigned reliability numbers; i.e., that the aircraft meets all the electrical and mechanical requirements to carry the nuclear weapon, and the environments that the nuclear weapon will experience while on the aircraft are within those described in

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the current nuclear weapon STS. The MAR is an NNSA statement that WR weapon material is satisfactory for release to the DoD for specified capabilities and uses. It is prepared by SNL and the appropriate physics laboratory (either Los Alamos National Laboratory or Lawrence Livermore National Laboratory) and approved by DOE/NNSA. The MAR may be qualified by limitations and exceptions. The MAR is published by NNSA and distributed to the DoD in a joint service technical manual. The MAR is a NNSA statement that WR weapon material is satisfactory for release to the DoD for specified capabilities and uses.

For air launched cruise missiles carried on aircraft, the certification tasks are similar to aircraft certification tasks described in the previous section; a CC is released by SNL for air-launched cruise missiles since the warhead interface is with the missile. Interface Control Drawings/Documents (ICDs) are developed, or updated, to define electrical and mechanical design requirements between aircraft and missile, and the missile and warhead. An EICD and MICD as defined in MIL-STD-1822 are also required by AFNWC and SNL to define the electrical and mechanical design requirements between the aircraft and the missile. Interface requirements must be established early in the program so that DOE/NNSA requirements are reflected throughout the weapon system. It is important to note that there can be changes to the aircraft/missile interface during the life of the weapon system that do not necessarily affect the warhead's MAR and CC. However, if there are changes to the warhead/missile interface, there is a good chance that changes may be required to the warhead's MAR and CC documents.

Each ACCD and CC is "owned" by the appropriate Sandia weapon engineering departments. When changes are required, either additions or deletions, the Sandia Aircraft Compatibility department prepares a memo to the appropriate weapon department outlining the changes and the rationale for those changes. They prepare the change paper and the Sandia Aircraft Compatibility department signs off on the final ACCD or CC.

5.19 Sign/approve/release Nuclear Compatibility Certification Statement (NCCS) (RA: AFNWC). The NCCS is published by the AFNWC when all aspects of the nuclear weapon system's compatibility certification have been completed. A NCCS is published for each aircraft capable of delivering nuclear weapons. The NCCS documents the nuclear weapon system configuration, carriage/delivery parameters, test information and references pertaining to compatibility of the delivery system with the nuclear weapon(s). Release of the NCCS constitutes nuclear compatibility certification of the delivery system with the specific weapon indicated; the NCCS is forwarded to the AFNWC certification branch as part of the overall nuclear certification package.

The NCCS is written as an unclassified document and is released in accordance with AFI 63-103, Nuclear Weapons Life Cycle Management. What it is, and how it fits in with the overall nuclear weapon system's certification process, is defined in AFI 63-125, Nuclear Certification Program. The following is a list of information to be included in each Section of the NCCS. Additional aircraft specific information, pertinent to nuclear weapon compatibility certification, may be provided in the NCCS.

Section I, Nuclear Aircraft System General Information, includes the aircraft nomenclature, the type of propulsion, what nuclear weapons it can carry, and what are the nuclear certified stations.

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Section II, AMAC Components, includes the hardware and software nomenclature, part/dash numbers, National Item Identification Number (NIIN), and Computer Program Identification Number (CPIN) identifying numbers.

Section III, Suspension and Release Equipment, includes pylons, rotary launchers, bomb rack nomenclature, part/dash and NIIN numbers, orifice size, ejection velocities, explosive cartridge configurations, and suitable substitute cartridges.

Section IV, Compatible Weapon/Equipment Configurations, includes weapon type, pull-out cable nomenclature, fin configuration, delay timer settings for parachute and spin rockets, configurations with nuclear weapons and other adjacent stores (fuel tanks, dispensers, pods, air-to-air missiles, etc.).

Section V, Carriage and Employment Limitations, includes weapon loadouts, min/max airspeed and altitudes.

Section VI, AMAC Testers, includes any pre-load and certification testers used to verify the AMAC interface on the aircraft before a nuclear bomb or missile is loaded on the aircraft. This includes the tester's nomenclature, part/dash numbers, CPIN, and NIIN numbers.

Section VII contains the following Appendices for specific reference information:

Appendix A, Historical AMAC Testing, covers both certification and surveillances tests. Dates of the tests, base location, aircraft tail numbers, and AMAC software version tested are listed. Results of the tests, including any Product Quality Deficiency Reports, and Conclusions to state compliance to the AMAC specifications are included for historical purposes.

Appendix B, Nuclear Certification Documents Reference, lists the titles of all applicable safety and compatibility documents and their date of release. This includes the SOC letters from Sandia, the Nuclear Safety Analysis Report (NSAR), the AMAC FDAR, the EICD and MICD, and applicable ACCD and CC. This is also a good location to list the versions of AMAC specifications that the aircraft was originally built to. This is helpful to know when AMAC surveillance testing finds discrepancies in test measurements compared to AMAC specification requirements. Because the current weapon stockpile has lighter electrical loads compared to weapon electrical loads of retired systems, the current System 1 specification reflects the current electrical loading in the stockpile.

Appendix C, Nuclear Certified Technical Orders, includes, but is not limited to, titles to all aircraft ground and in-flight Operating TOs, nuclear weapons aircraft mate and demate TOs, Aircrew delivery procedures (nuclear weapons and trainers), ejector rack, maintenance, and tester manuals.

Appendix D, Open Issues and Restrictions, defines any carriage restrictions, max load configurations, degradation of AMAC circuits found during surveillance testing, and any other noted limitation or finding.

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It is important to keep all of these Sections up-to-date with the latest information. It is good for the author of the NCCS to review all Sections prior to the signoff of a new version of the NCCS, to make sure there is nothing that is out-of-date and all information is current and all Sections are complete. In addition, in the preliminary stages of updating the NCCS, it is important to perform a cross-check of configuration information supplied by the System Program Manager, to the information in the MNCL. This helps the MNCL be an accurate representation of what is allowed in the field.

5.20 Publish Master Nuclear Certification List (MNCL) (RA: AFNWC). The MNCL is a domain controlled web-site data base that identifies equipment (hardware and software) that is nuclear certified per AFI 63-125. The MNCL is the sole authority for determining equipment certification status, and is managed by the AFNWC's certification management organization. After the NCCS has been published signifying the satisfactory completion of the nuclear certification process, the MNCL is updated. The NCCS is a source document for the MNCL; therefore, this update to the MNCL is based on software, hardware, and support equipment items listed in the NCCS.

5.21 Compatibility certification activities/ground launched missile (RA: AFNWC, SA: NNSA/SNL). The MC for a ground launched missile warhead is a DoD prepared document submitted to the NNSA that specifies performance requirements and physical characteristics for a nuclear warhead to be compatible with a specific weapon system or systems. A STS is the DoD document that defines the logistical and employment concepts and related normal and abnormal environments, including vulnerability criteria, involved in the delivery of a nuclear weapon from the stockpile to the target. It may also define the logistical flow involved in moving nuclear weapons to and from the stockpile for quality assurance testing, modification and retrofit, and the recycling of limited-life components. ICDs are developed or updated, as appropriate, by the DoD to define electrical and mechanical requirements between the delivery platform (missile) and the warhead. ICD interface requirements must be defined jointly by DoD and NNSA early in the program and updated as appropriate so that NNSA interface requirements are reflected throughout the weapon system. A MAR for a ground launched missile is a NNSA approved statement that war reserve (WR) weapon material is satisfactory for release on a designated effective date to the DoD for specified uses which are qualified by exceptions and limitations. The MAR identifies the applicable requirements documents (MCs and STS) and ICDs. The MAR is prepared by Sandia National Laboratories and approved by NNSA, SNL, and either Los Alamos National Laboratory or Lawrence Livermore National Laboratory. Issuance of the MAR indicates the completion of the missile/warhead compatibility analysis and documentation required for nuclear certification. It must be emphasized that changes to the warhead/missile interface can directly impact the MC, STS, ICDs and MAR. Issuance of the STS and MC by the DoD, the MAR by NNSA, as well as the jointly defined NNSA/DoD ICDs, culminates the missile/warhead compatibility analysis and documentation required for nuclear certification.

5.22 Establish Resident Compatibility Technical Basis (RA: AFNWC, SM, SA: NNSA/SNL). Department of Energy compatibility recommendations are based on extensive engineering analysis using data obtained from tests identified earlier in this section. The data used in this analysis form the technical basis upon which engineering judgement is obtained to render advice to the nuclear weapons program managers to include modified weapon/aircraft or new weapon/

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aircraft in the appropriate compatibility configuration drawings. Periodically, this technical basis must be revisited to examine how a decision was previously made or other reasons. As such, all data used in making compatibility determinations must reside within DOE facilities. Aircraft platform program offices, in consultation with AFNWC, must ensure Sandia National Laboratories is able to obtain and store data in DOE facilities from tests associated with compatibility certification activities in order to form the technical basis necessary to issue decisions and advice.

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6. NOTES

6.1 Intended Use. This military handbook provides guidance and information to DoD, NNSA, and other individuals, who are assigned responsibility for aircraft compatibility nuclear certification. Its purpose is to assist them in planning for and implementing and sustaining nuclear certification activities. It supports specifications, standards and Air Force Instructions to the greatest practicable extent.

6.2 Key Word Listing.

a. Aircraft Monitor and Control

b. AMAC Simulators

c. AMAC Specifications

d. AMAC testing

e. Weapon Separation

f. Internal Carriage

g. External Carriage

h. Vibration Fly-around

i. Static Ejection

j. Mechanical Interface

k. Electrical Interface

l. Mechanical fit

m. Preliminary Design

n. Final Design

o. Five Aircraft Test

p. Compatibility Test Unit

q. Compatibility Certification

r. Cruise Combat Load

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s. Simulated Flight

t. Electromagnetic Test

u. Full Weapon System Drop Test

v. Statement of Compatibility

w. Aircraft Compatibility Control Drawing

x. Major Assembly Release

y. Military Characteristics

z. Nuclear Compatibility Certification Statement

aa. Nuclear Certification Program

ab. Nuclear Safety

ac. Master Nuclear Certification List

ad. Stockpile-to-Target Sequence

6.3 Changes from Previous Issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

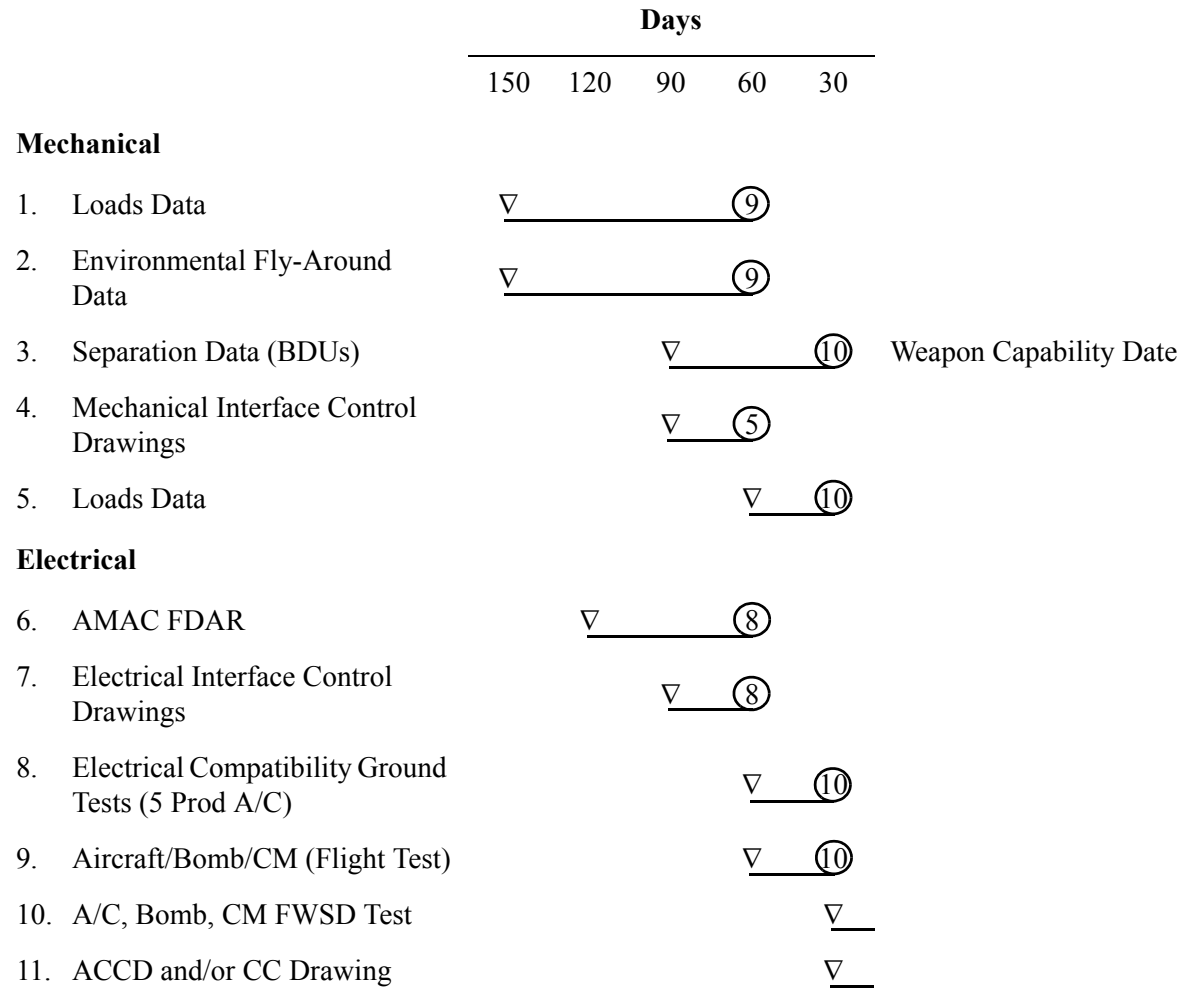
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MINIMUM TIME NEEDED TO COMPLETE WEAPON CAPABILITY DATE TASKS

A.1 SCOPE

A.1.1 Scope. Appendix A contains a chart showing what historically has been the minimum time needed by Sandia to complete the tasks necessary to get to a Weapon Capability Date (WCD). Minimum times should only be used when time is of the essence in order to meet a WCD. The WCD is the date the user says they can carry nuclear bombs on the aircraft. Sometimes the WCD is called a Rules Need Date (RND). The ACCD/CC and NCCS must be released prior to or in conjunction with the WCD.

TABLE A-I. Minimum time to weapon capability date



code: ∇ - Need Date
— - Time for Analysis
◯ - Next Event

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OUTLINE OF A NUCLEAR BOMB/CRUISE MISSILE COMPATIBILITY TEST PROGRAM

B.1 SCOPE

B.1.1 Scope. Appendix B contains an outline of a nuclear weapon compatibility test program. The Tables in this Appendix show who is responsible for accomplishing the many tasks necessary to grant a nuclear capability in an aircraft program. It outlines the following:

a. What the item is,

b. What is needed and by who,

c. What data is to be obtained,

d. Who does the analysis, and

e. What other requirements are needed for each specific item.

These charts are helpful when explaining to Sandia weapon departments, the DoD, NATO, and the aircraft contractor what the level of effort is and who is expected to participate to have a capability with a nuclear weapon on an aircraft.

B.1.2 Ground Tests.

a. Wind Tunnel

b. Static Drops

c. Mechanical Fit

d. Electrical Function

B.1.3 Flyaround Tests.

a. Lug and Sway Brace Loads

b. Vibration

c. Thermal

d. Electrical Function

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B.1.4 Air Drop Tests.

a. Dynamic Drops

b. Full Weapon System Drops

TABLE B-I. Ground tests

	ITEM	NEEDED	DATA TO BE OBTAINED	ANALYSIS BY	OTHER REQUIREMENTS
1.	Wind Tunnel	US or C Wind Tunnel US or C Aircraft & Pylon Models US or SNL Bomb/CM Models	Loads Data on Bombs/CMs	US, SNL & C	US Config and Mixes
2.	Static Drops	US or C Ejector Rack US or C Test Facility US or C Weighted & cg Test Units	Pitch Ejection Velocity Recommended Cart & Orifice	US or C	US or C Data to be Integrated into Wind Tunnel Separations and Used Later in Air Drop Test Program
3.	Mechanical Fit	a) C Mock-up or Pre-Prod C or SNL/US Shapes (BDU's or CTUs or CMs) SNL System 1 Cables US System 2 Cables	Clearances and Serviceability	US, SNL & C	C Mech Layouts or Prelim Mech Interface Control Drawing (MICD) and 5 A/C AMAC test
		b) US or C Prod A/C SNL CTUs & System 1 Cables US CMs, System 2 Cables, Load, Checkout Equip and Crew	Verification of Fit Plus: AF-16 T.O.'s	US, SNL & C	US or C Completed Mech Interface Control Drawing (MICD) AF or C - 16 T.O.'s (Blue Line)

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TABLE B-I. Ground tests - Continued

	ITEM	NEEDED	DATA TO BE OBTAINED	ANALYSIS BY	OTHER REQUIREMENTS
4.	Electrical Function of EM Equip on A/C (may not be required depending on adequacy of data supplied)	US or C Prod A/C with EM Equip SNL EMMS or EMC units & System 1 Cables US CMs, System 2 Cables, Load, Checkout Equip and Crew	EM exposure to Bombs/CMs	SNL	SNL will Perform an Analysis if a problem exists that would impact the weapon's compatibility with the aircraft's radiating equipment
5.	Electrical Function of AMAC	C Mock-up or Pre-Prod A/C SNL CTUs & System 1 Cables US CMs & System 2 Cables AF & SNL Record & Test Equip	Preliminary Verification of AMAC Spec and ICD Verify Quality of Elec Signals to Bombs/CMs	US, AF, SNL & C	US or C Prelim Submittal of Preliminary Design Report (PDR)
		US or C Five Prod A/C SNL CTUs & System 1 Cables US CMs, System 2 Cables, Load, Checkout Equip & Crew AF & SNL Record & Test Equip	Verify Compliance with Appropriate AMAC Specs and Quality of Elec Signals Between Bomb/CM and A/C Plus: AF-16 T.O.'s	US, AF, SNL & C	US or C Final Design Approval Report (FDAR), Final Mech and Elec Interface Control Drawings (MICD/EICD) AF or C-16 T.O.'s (Blue Line)

Code: C = Aircraft Contractor; SNL = Sandia National Labs; US = Using Service (USAF or NATO); AF = U.S. Air Force

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TABLE B-II. Flyaround tests

	ITEM	NEEDED	DATA TO BE OBTAINED	ANALYSIS BY	OTHER REQUIREMENTS
1.	Lug and Sway Brace Loads	If Table B-I, Item 1, needs to be verified SNL test unit or C instrumented bomb rack required			
		For Additional Needs Same as Table B-II item 2 Vibration	Lug and Sway Brace Reactions at Various Flight Conditions and Maneuvers	SNL	Same Requirements as in Vibration
		US or C Best Results Can Be Obtained with an "Instrumented" Aircraft for Data Correlation			
2.	Vibration	SNL Units T/M			
		US or C Aero and Struct Equivalent Aircraft			US or C Will Provide A/C Flight Clearance
		US or C Method to Control VFA Unit from Cockpit	Sample of the Vibration Environment of Certain Components Within Bomb/ CM During Specified Flight Conditions	SNL	SNL May require Wind Tunnel Loads Data and Aircraft Post Flight Performance Data Tapes
		SNL Ground Station for Pre- Flight Checkout			SNL Will Perform an Analysis to Determine Bomb Structural Adequacy Prior to These Flights
		US or SNL Flight Test Area			
		US or C CM, Load Checkout Equip & Crew			

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TABLE B-II. Flyaround tests - Continued

	ITEM	NEEDED	DATA TO BE OBTAINED	ANALYSIS BY	OTHER REQUIREMENTS
3.	Thermal (may not be required depending on adequacy of data supplied)	US or C A Definition of Thermal Environment Imposed on Bombs	Data Sufficient to Create a Thermal Model of Weapons Bay with Doors Closed & Opened or on Pylon Superimpose Bombs/CMs in This Thermal Environment	SNL	Same requirements as in Vibration
4.	Electrical Function of EM Equip on A/C (may not be required depending on adequacy of data supplied)	US or C Prod A/C with EM Equip SNL EMMS or EMC units & System 1 Cables US CM, System 2 Cables, Load Equip Checkout and Crew US or C Production Aircraft	EM exposure to Bombs/CMs	SNL	Same Requirements as in Table B.1 Item 4 Electrical
5.	Electrical Function of AMAC	SNL CTUs and System 1 Cables US & C CMs, System 2 Cables, Load Equip Checkout & Crew Production Aircraft AF & SNL Record & Test Equip	Observe Correct Operation Thru Prearm During Flights in Local Area Verify Proper Quality of Elec Signal to WPN	US, SNL & C	Same Requirements as in Electrical Table B.1 Item 5

Code: C = Aircraft Contractor; SNL = Sandia National Labs; US = Using Service (USAF or NATO); AF = U.S. Air Force

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TABLE B-III. Air drop tests

	ITEM	NEEDED	DATA TO BE OBTAINED	ANALYSIS BY	OTHER REQUIREMENTS
1.	Dynamic Drops	US or C BDU's or SNL FTU's US or C Aircraft with On-Board Cameras US or C Photo Chase and/or GND Cameras US or C Test Plan Based on Wind Tunnel Separation Tests	Full Scale Demonstrations Throughout Flight Envelope Desired by User Free Fall and Retarded Options	US, SNL & C	US or C Will Provide Flight Clearance and Decide on Progression to Next Data Point
2.	Full Weapon System Drops	US or SNL WPNs & Cables US Production Aircraft SNL or US Test Range with Ability to Determine Trajectories US CM, Load, Checkout Equip and Crew	Complete Weapon System Functional Tests, includes Mission Planning Separation Ballistic Elec Interface AF&F All Options	US & SNL	US or C Will Provide Flight Clearance

C = Aircraft Contractor; SNL = Sandia National Labs; US = Using Service (USAF or NATO); AF = U.S. Air Force

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ELECTRICAL COMPATIBILITY TASKS CHECKLIST

C.1 SCOPE

C.1.1 Scope. Appendix C provides a checklist to aid the AFNWC and Sandia engineers to determine when and what kind of AMAC electrical tests are required based upon the type of changes made to the aircraft platform. Similar data can be found in the AMAC Specification documents.

TABLE C-I. Checklist for electrical compatibility tasks

AIRCRAFT CHANGES (AMAC Electrical, Hardware, & Software)	AMAC TESTS AND ANALYSES TO BE COMPLETED				
	System 1 or 2 Specification Compliance	Ground Functional of CTUs	Flight Functional of CTUs	Full Weapon System Drop	Engineering Analysis
New Aircraft System or New Aircraft Variant	X	1, 2	1, 2	5	X
Major Hardware Modifications with Interface Impact	X	1, 2	1, 2	(5)	X
Minor Hardware Modifications with Interface Impact	(X)	1, 2			X
Hardware Modifications with No Interface Impact	(X)				X
AMAC Electrical Power Modification	X	1, 2	1, 2	(5)	X
Major Software Modifications with Interface Impact	X	1, 2	1, 2	(5)	X

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TABLE C-I. Checklist for electrical compatibility tasks - Continued

AIRCRAFT CHANGES (AMAC Electrical, Hardware, & Software)	AMAC TESTS AND ANALYSES TO BE COMPLETED				
	System 1 or 2 Specification Compliance	Ground Functional of CTUs	Flight Functional of CTUs	Full Weapon System Drop	Engineering Analysis
Minor Software Modifications with Interface Impact	X	1, 2, 3, 4			X
Software Modifications with No Interface Impact	(X)	(1, 2, 3, 4)			X
No Changes to AMAC System for 5 - 10 years	X	1, 2	1, 2		X
New or LEP wpn	X	1, 2	1, 2	5	X
Major Electrical Changes that Impact the Interface	X	1, 2	1, 2		X
Minor Electrical Changes that Impact the Interface	(X)	1, 2	1, 2		X
Electrical Change that does not Impact the Weapon Interface	(X)	(1, 2, 3, 4)			X

LEGEND:

( )	Test is optional - To be required on a case-by-case basis
X	No Compatibility Test Units Required
1	CTU-1
2	CTU-2
3	CTU-3
4	CTU-4
5	Full Weapon System Drop Test Unit

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MECHANICAL COMPATIBILITY TASKS CHECKLIST

D.1 SCOPE

D.1.1 Scope. Attachment D provides a checklist to aid the AFNWC and Sandia engineers to determine when and what kind of mechanical tests are required based upon the type of changes made to the aircraft platform.

TABLE D-I. Checklist for mechanical compatibility tasks

AIRCRAFT CHANGES (Mechanical Hardware)	MECHANICAL TESTS AND ANALYSES TO BE COMPLETED				
	Wind Tunnel	**Static Drops**	**Mechanical Fit**	**Full Weapon System Drop**	**Engineering Analysis**
New Aircraft System; New Aircraft Variant	6, 7	8	1, 2	5	X
Major Hardware Modifications with Interface Impact	6, 7	8	1, 2	(5)	X
Minor Hardware Modifications with Interface Impact	(6, 7)	(8)	(1, 2)		X
Hardware Modifications with No Interface Impact	(6, 7)	(8)			X
New Weapon or LEP Weapon	6, 7	8	1, 2	5	X

LEGEND:		
	( )	Test is optional - to be required on case-by-case basis
	X	No Compatibility Test Units Required
	1	CTU-1
	2	CTU-2
	3	CTU-3
	4	CTU-4
	5	Full Weapon System Drop Test Unit
	6	Aircraft Model
	7	Bomb/CM Model
	8	BDU

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CONCLUDING MATERIAL

Custodians:

Army - AV
Air Force - 27
Navy - AS

Preparing Activity:

Air Force - 27
(Project 11GP-2016-001)

Review Activities:
Air Force - 11
DTRA - DS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at: https://assist.dla.mil.

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